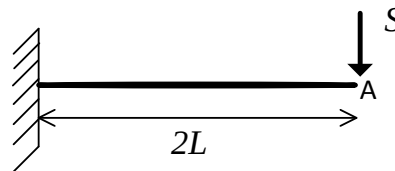


5 (long) Figure 5 shows a set of uniform beams, all with flexural rigidity EI .

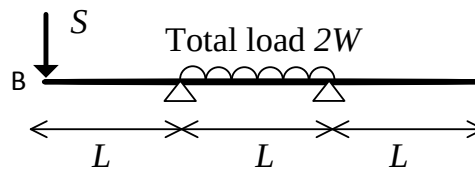
(a) The beam shown in Fig. 5a is subjected to a downwards vertical load S at A. Calculate the vertical deflection of point A. [5]

(b) The beam shown in Fig. 5b is subjected to a downwards vertical load S at B and a uniformly distributed load over the central span. Calculate the vertical deflection of point B. [20]

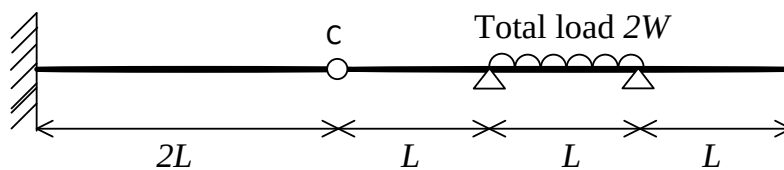
(c) For the beam shown in Fig. 5c, calculate the vertical deflection of point C. [5]



(a)



(b)



(c)

Fig. 5